

Zeve J. Marcus

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Summary

Researcher and Ph.D. Candidate (Experimental Psychology) with extensive experience in statistical modeling, machine learning, natural language processing, psychometrics and experimental design. Adept at combining rigorous academic research methods with industry-focused analytics to drive actionable insights. Recognized for automating data workflows, conducting multilevel and Bayesian analyses, delivering clear visualizations and data-driven recommendations to diverse audiences.

Education

Ph.D., Experimental Psychology (Expected June 2025) University of California, San Diego Research Focus: Experimental Evolutionary Psychology, advanced quantitative methods	2025
M.A., Experimental Psychology University of California, San Diego Research Focus: Cognition, Perception and Cognitive Neuroscience <i>Thesis: The Influences of State-Change Expectations on Change Detection in Natural Scenes</i>	2020
B.A., Psychology University of California, San Diego	2016

Skillset

Programming & Analysis: R, Python, SQL (Advanced), SPSS, MATLAB
Statistical & Data Methods: Bayesian Regression (BRMS/Stán), SEM, Multilevel Modeling, Machine Learning (clustering, predictive modeling), NLP, Survey Design
Data Visualization & BI Tools: Power BI, SSMS, Qualtrics, Sogolytics
Research & Soft Skills: Experimental Design, Mixed Methods Research, Psychometrics, Communication, Teaching & Mentoring

Professional Experience

Research Fellow & Statistical Consultant (Contract) Collective Action Research Emergence (CARE) Institute	2024
- Designed and implemented advanced statistical workflows in R and Python to analyze large-scale, survey-based political ideology data. - Applied clustering techniques, density-based thresholding, and Gower’s distance methods to segment populations and uncover behavioral insights.	

- Communicated findings via executive summaries and data visualizations for academic and policy-making audiences.

Research Specialist

2024

TI Verbatim Consulting

- Applied advanced statistical modeling to identify key drivers of organizational culture, climate, and effectiveness.
- Led Python-based data analysis pipelines to detect complex trends in interview and focus group data, producing comprehensive reports that informed key decision-making processes.
- Developed advanced ML/NLP approaches for qualitative data:
 - Implemented LDA and multidimensional scaling for topic modeling and dimensionality reduction.
 - Utilized BERT and HDBSCAN to identify nuanced topic clusters, revealing latent themes in large text corpora.
 - Conducted sentiment analysis with BART to classify sentiments accurately.
 - Deployed RoBERTa-based emotion detection for granular emotional insights, leveraging dynamic weighting techniques to improve model precision.
 - Integrated models to correlate survey data, sentiments, and emotions at the individual level, providing holistic insights for clients.
- Created intuitive visualizations for exploratory analyses and client-facing presentations.
- Led design and guided implementation of a web-based data analysis tool incorporating quantitative and qualitative ML methods.
- Designed and programmed surveys using robust psychometric evaluations (EFA, CFA) to ensure validity and reliability of constructs.
- Developed interactive R Shiny applications for collaborative insights:
 - Integrated linear regression models with automated threshold-based survey scoring systems, dynamically updating outputs by demographic variables.
 - Enabled efficient exploration of complex datasets via user-friendly dashboards and data-drilldown features.
- Conducted thorough literature reviews and compiled whitepapers to contextualize findings.

Sr. CX Insights Analyst (Temporary)

2022-2023

DentaQuest

- Automated data processing workflows using Python and SQL, reducing turnaround time by 30% for report delivery.
- Developed and deployed NLP and machine learning models to perform topic and sentiment analysis on verbatim customer feedback.
- Created Power BI dashboards that identified improvement opportunities for outreach and customer journey, influencing strategic decisions.
- Conducted journey mapping to understand dental providers' office-staff experiences, enabling targeted interventions.

Research Experience

Graduate Student Researcher*2020-2025*

The Evolution and Human Behavior Lab, University of California, San Diego

- Led analyses of large datasets (public and lab-collected) using Bayesian regression, CFA/SEM, and multilevel modeling to investigate attention, prosocial decisions, and cultural norms.
- Programmed and conducted Tobii-based eye-tracking experiments and Python-based clickstream data collection for economic decision-making studies.
- Designed surveys (Qualtrics, Java integrations) to measure behavior, beliefs, and prosocial attitudes; assessed psychometric properties of novel instruments.
- Authored and co-authored peer-reviewed manuscripts and presented findings at conferences.

Graduate Student Researcher (Contract)*2019-2022*

The Lipomi Research Group in the Department of Nanoengineering,
University of California, San Diego

- Provided biostatistics consultation and performed power analyses for psychophysics experiments on haptic stimuli.

Graduate Student Researcher*2018-2019*

Ramachandran Lab

The Center for Brain and Cognition - University of California, San Diego

- Designed and implemented MATLAB- and Python-based experiments examining cognition and attention.
- Performed statistical analysis (R) and data visualization, co-authoring manuscripts in neuroscience journals.

Graduate Student Researcher*2017-2018*

Evolutionary Psychology Lab

Center for Evolutionary Psychological Science - Oakland University (Michigan)

- Planned research questions and designed experiments on evolutionary factors in social and cognitive processes.
- Collected and analyzed survey data using R.

***Additional Early Research Roles:** Post-Baccalaureate Research Assistant at Ramachandran Lab (UC San Diego, 2016–2018) and Undergraduate Research Assistant (2014–2015). Responsibilities included designing experiments, training new lab members, coordinating data collection, conducting literature reviews, publishing in peer reviewed journals.

Teaching and Mentoring**Career Transition Representative***2023-2024*

Psychology Program, University of California, San Diego

- Organized an annual career symposium, inviting industry and academic speakers; led panels to facilitate career guidance for graduate students.

- Created an online resource hub (blog, wiki, Google Drive) to connect grads with relevant career services.

Learning Facilitator (Lecturer)

2021-2022

UCSD Extension

- Primary Instructor
 - Introduction to Psychology

Graduate Teaching Assistant

2018-2023

University of California, San Diego

- Led discussion sections, created teaching materials, graded assignments and exams, held office hours for 200–300 students, and delivered occasional guest lectures.

Selected courses:

- PSYC 161 - Engineering Psychology
- PSYC 148 - Psychology/Judgement & Decision-Making
- PSYC 114 - Lab/Psychophysiological Perspectives on the Social Mind
- PSYC 144 - Memory and Amnesia
- PSYC 60 – Introduction to Statistics
- PSYC 124 – Clinical Neuropsychology

Guest Lectures:

Attention and Prosocial Behavior (PSYC 114 - Lab/Psychophysiological Perspectives on the Social Mind)

Early Development of Morality (PSYC 190 - Science of Parenting)

Neuro-Biology of Stress (PSYC 2 - Gen Psych-Biological Foundations)

Cognitive Therapy Approaches (PSYC 124 – Clinical Psychology)

Traumatic Brain Injury (PSYC 124 – Clinical Neuropsychology)

Selected Publications

Marcus, Z. J. & McCullough, M. E. (2020) Does Religion Make People More Self-Controlled? A Review of Research from The Lab and Life. *Current Opinion in Psychology*
doi:[10.1016/j.copsyc.2020.12.001](https://doi.org/10.1016/j.copsyc.2020.12.001).

Ramachandran, V.S., **Marcus, Z.J.**, Chunharas, C., (2020) Bouba-Kiki: Cross-domain resonance and the origins of Synaesthesia, Metaphor, and Words in the Human Mind. *Multisensory Perception*. Elsevier.

Ramachandran V.S., Chunharas C., & **Marcus Z.** (2018). Constructing Calendars in the Brain. *Neurocase*. 2020 Feb;26(1):7-17. doi: 10.1080/13554794.2019.1694040.

Ramachandran V.S., Chunharas C., **Marcus Z.**, Furnish T., Lin A. (2018). Relief From Intractable Phantom Pain by Combining Psilocybin and Mirror Visual Feedback (MVF). *Neurocase*. 2018 Apr;24(2):105-110. doi: 10.1080/13554794.2018.1468469

Ramachandran, V. S., & **Marcus, Z.** (2017). Synesthesia and the McCollough Effect. *Perception*. 2017 Jun 1;8(3):2041669517711718. doi: 10.1177/2041669517711718.

(Complete list available upon request)

Selected Conference Presentations

Attention and Prosocial Decision Making in Economic Games Marcus Z.J, E. Pedersen, M.E. McCullough <i>The Society for Personality and Social Psychology</i>	2022
Memory Consolidation in Sound-Form Symbolism (Bouba-Kiki) and Word-Color Concordances Ramachandran V.S, Bhattacharya P., Marcus Z.J., Sen R., Vonkliest K., Chunharas C. <i>Society for Neuroscience</i>	2020
The Neural Basis of Mental Calendars Ramachandran V.S, Marcus Z.J. <i>Society for Neuroscience</i>	2020
Subtle Emotional Distinctions in Primate Vocalizations Can Be Recognized By Naive Humans Ramachandran V.S, Kandalaft S., Vayyala K., Chunharas C., Marcus Z.J <i>Society for Neuroscience</i>	2018

(Complete list available upon request)

Additional Training

- **Introduction to Applied Biostatistics** (edX)
 - **Survival Analysis in R** (DataCamp)
 - **Bayesian Statistics in R** (DataCamp)
 - **Behavioral Economics in Action** (edX)
 - **Workshop Series on Teaching & Mentoring in Psychology** (UC San Diego, 2021)
 - **Diversity, Equity, and Inclusion Workshop** (UC San Diego, 2021)
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Community Involvement

Mesa College's UCSD Psychology Graduate Student Panel Meet n' Greet <i>Panelist/Organizer</i>	2020
UCSD Experimental Psychology Graduate 2023 Departmental Career Symposium <i>Panelist</i>	2023

Organizations

The Society for Personality and Social Psychology	2021-present
American Psychological Association (APA)	2020-present
Psychonomic Society	2019-present
Society for Neuroscience	2018-present